William Schober

PhD student in Quantum Computing

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I am a PhD student in the Cryptography and Quantum Information group at the Università della Svizzera italiana, under the supervision of Stefan Wolf. My research focuses on developing mathematical frameworks for manipulating quantum circuits via local circuit equivalences. I am motivated by improving the suite of local circuit optimization techniques employed by quantum compilers, variously called *peephole optimizations* or *template optimizations*.

I expect to defend my thesis in August 2027. I am currently looking for work starting from September 2027. My ideal position would be to join the compiler team in a quantum computing company, though I am open to similar postdoc positions. I have a preference for institutions that are developing their own hardware, and for working environments that are highly collaborative, friendly, and results-driven.

Education

2022 - 2027 **PhD in Informatics**, Università della Svizzera italiana, Lugano, Switzerland
Thesis: *Equational theories for quantum circuits, with application to circuit optimization.* Supervisor: Prof. Stefan Wolf

2019 - 2021 **MSc in Artifical Intelligence**, Università della Svizzera italiana, Lugano, Switzerland
Thesis: *Geometry of quantum state space: an investigation by convexity and unitary orbits.* Supervisor: Prof. Stefan Wolf

2014 - 2017 **BSc in Physics and Mathematics**, University of Massachusetts Amherst, Amherst, MA, USA

Undergraduate Research: Analyzed collision data from the Large Hadron Collider using binary classification neural networks written in ROOT, a C++ derived programming language, on a Linux platform. Supervisor: Prof. Stéphane Willocq

Publications

2026 **W. Schober** and S. Wesley. A New Method For Manipulating Circuits, Application To Quantum Adders.
arXiv

2026 **W. Schober** and S. Wesley. A Complete Equational Theory for Quantum Circuits with Generalized Control. *Electronic Proceedings in Theoretical Computer Science* 456, pp. 283-349.
Best Student Paper Award at the 23rd International Conference on Quantum Physics and Logic (QPL 2026).
arXiv DOI

2024 **W. Schober**. Extended Quantum Circuit Diagrams. arXiv

Teaching and Supervision

2022 - 2026

Teaching assistant, Computer Science courses, Università della Svizzera italiana

Led exercise sessions and designed and graded homework and exams, working in teams of 1-4 TAs for classes of 20-80 students.

Information & Physics	MSc course	Spring 2026
Software Atelier 1: Fundamentals of Informatics	BSc course	Fall 2025
Information & Physics	MSc course	Spring 2025
Software Atelier 1: Fundamentals of Informatics	BSc course	Fall 2024
Information & Physics	MSc course	Spring 2024
Programming Fundamentals 1	BSc course	Fall 2023
Information & Physics	MSc course	Spring 2023
Numerical Algorithms	MSc course	Fall 2022

2025 - 2026

Thesis supervisor, Computer Science students, Università della Svizzera italiana

- **Samuele Galbusera** BSc thesis, 2026
  Does provable absence of barren plateaus imply classical simulability?
- **Mark Sobolev** MSc thesis, 2025
  On Qubit Routing in Quantum Computer Architectures.

Talks

Oct 2026

A Complete Equational Theory for Quantum Circuits with Generalized Control. Seminar talk, École de technologie supérieure quantum seminar series, Quebec. Video

Aug 2026

A Complete Equational Theory for Quantum Circuits with Generalized Control. Conference talk, QPL 2026, Amsterdam. Video

Jan 2026

Towards a Hierarchical Quantum Circuit Language. Workshop talk, PLanQC 2026, Rennes. Video

Posters

Feb 2025

W. Schober. *Extended quantum circuit diagrams.* QIP 2025, Raleigh.

Sep 2024

W. Schober. *Circuits as a self-contained graphical language.*TQC 2024, Okinawa.

Jul 2024

W. Schober. *Circuits as a self-contained graphical language.* QPL 2024, Buenos Aires.

Jan 2024

W. Schober. *Extended Quantum Circuit Diagrams.* QIP 2024, Taipei.

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